

# FATIMA ABOUL

Software Developer | Computer Science Student

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

## SUMMARY

Motivated final-year Computer Science student with hands-on experience in software development, AI-powered applications, full-stack web development, and database-driven systems. Skilled in Python, JavaScript, React, Node.js, C#, and SQL, with a growing focus on modern AI integration. Eager to contribute to real-world software projects while continuing to grow as a developer.

## EDUCATION

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|----------------------------------------------------------------------------------------|---------------------|
| <b>Bachelor of Science in Computer Science</b><br><i>DHA Suffa University, Karachi</i> | Oct 2023 – Present  |
| <b>Intermediate (Pre-Engineering)</b><br><i>TCF College, Qayyumabad, Karachi</i>       | Aug 2021 – Jul 2023 |

## EXPERIENCE

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| <b>Tutor — FAHAD Tutors FTA</b><br><ul style="list-style-type: none"><li>Tutored students of different age groups and skill levels.</li><li>Strengthened communication, writing, and interpersonal skills.</li></ul>                                                                  | 2024 – Present |
| <b>Volunteer — Climate Change Initiative</b><br><ul style="list-style-type: none"><li>Participated in environmental awareness and community engagement activities.</li><li>Developed a stronger sense of social responsibility and teamwork.</li></ul>                                | Jan 2023       |
| <b>Volunteer &amp; Ambassador — The Citizens Foundation (TCF)</b><br><ul style="list-style-type: none"><li>Proctored examinations with integrity in a fair, distraction-free environment.</li><li>Guided students and provided information about TCF programs and services.</li></ul> | 2024 – Present |

## TECHNICAL SKILLS

**Programming Languages:** Python, C#, JavaScript, SQL  
**Web Technologies:** HTML5, CSS3, React, Node.js, Express.js  
**AI / ML:** Google Gemini API, RAG, Embeddings, Streamlit  
**Database & APIs:** MySQL, REST APIs, JWT Authentication  
**Tools & Version Control:** Git, GitHub, Visual Studio, VS Code, MS Word, PowerPoint

## SOFT SKILLS

Communication, Teamwork, Leadership, Time Management, Problem Solving, Adaptability

## PROJECTS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>Flavr — Full-Stack Recipe Discovery App</b><br><i>React, Node.js, Express.js, MySQL</i> <ul style="list-style-type: none"><li>Developed a full-stack recipe discovery platform with user authentication and recipe browsing.</li><li>Implemented favorites, weekly meal planning, image uploads, and dark mode.</li><li>Built an admin dashboard for recipe management and content control.</li><li>Integrated frontend, backend APIs, and MySQL database for efficient data management.</li></ul> |
| <b>AI Agent</b><br><i>Python, Streamlit, Google Gemini API</i> <ul style="list-style-type: none"><li>Built an AI agent using Gemini API and function calling to handle user requests.</li><li>Integrated tools for calculations, web search, date/time lookup, and live currency conversion.</li><li>Implemented multilingual responses with persistent chat history and a custom Streamlit interface.</li><li>Added user-friendly error handling for reliable interaction.</li></ul>                 |
| <b>Retrieval-Augmented Generation (RAG) PDF Q&amp;A Application</b>                                                                                                                                                                                                                                                                                                                                                                                                                                   |

*Python, Streamlit, Google Gemini API*

- Developed a RAG-based PDF question-answering application using Python, Streamlit, and Google Gemini API.
- Implemented text extraction, chunking, semantic embeddings, and cosine-similarity-based retrieval using pypdf and sentence-transformers.
- Deployed a live AI application that generates context-grounded answers from uploaded PDF documents.

### **Student Management System**

*C# Windows Forms, MySQL*

- Developed a desktop application for managing student records.
- Implemented CRUD operations for adding, viewing, updating, searching, and deleting records.
- Added input validation to ensure accurate student data.
- Integrated MySQL database for reliable data storage and retrieval.

### **To-Do List App**

*HTML5, CSS3, JavaScript*

- Developed a responsive task-management web application with an interactive user interface.
- Implemented task filtering, important-task marking, and localStorage persistence.
- Added calendar and task statistics views for better task organization.
- Integrated profile settings and dark mode for an enhanced user experience.